

SHEPPARTON, Australia

WMO: 948750

Lat: 36.4289S

Lon: 145.3947E

Elev: 115

StdP: 99.96

Time Zone: 10.00 (AUS)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-0.6	0.6	-2.7	3.1	17.6	-1.2	3.5	12.2	9.9	12.1	9.0	11.6	0.5	160	0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.6	37.6	19.0	35.1	18.4	32.9	18.2	21.8	29.0	20.8	28.7	19.9	28.8	5.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	15.0	23.4	18.6	13.6	23.1	17.2	12.4	22.0	64.3	28.8	60.5	28.4	57.3	28.8	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.0	8.9	8.0	DB	-2.7	41.3	0.9	1.6	-3.4	42.4	-3.9	43.4	-4.4	44.3	-5.1	45.4	(4)
			WB	-2.8	23.4	1.0	1.0	-3.5	24.1	-4.1	24.7	-4.6	25.3	-5.3	26.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.3	23.3	22.6	19.8	15.4	11.4	8.8	9.1	11.5	14.6	18.3	20.7		
		DBStd	6.15	3.98	3.52	3.56	3.07	2.52	2.20	2.07	2.16	2.75	3.38	3.88		
		HDD10.0	181	0	0	0	1	13	48	59	44	13	2	0		
		HDD18.3	1621	4	5	23	96	214	286	310	287	205	126	46		
		CDD10.0	2111	412	353	306	164	58	12	8	16	58	144	248		
		CDD18.3	508	158	125	71	9	0	0	0	0	1	9	44		
		CDH23.3	6800	2136	1486	826	128	1	0	0	0	21	211	691		
		CDH26.7	3223	1166	722	331	19	0	0	0	4	68	289	625		
(14)	Wind	WSAvg	3.7	4.6	4.2	3.8	3.1	2.9	2.8	3.1	3.4	3.7	3.9	4.3	4.5	(14)
(15)	Precipitation	PrecAvg	504	37	28	30	41	48	44	51	51	51	49	37	38	(15)
(16)		PrecMax	894	126	122	110	152	125	142	122	112	129	180	104	132	(16)
(17)		PrecMin	216	0	0	0	1	2	5	9	5	9	5	0	1	(17)
(18)		PrecStd	154	33	32	28	33	33	28	26	26	28	40	27	34	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	41.4	39.6	36.3	29.1	22.5	17.8	16.8	19.9	26.0	32.2	35.7	39.2		
(20)		0.4% MCWB	19.3	20.2	18.9	16.3	14.2	12.4	11.9	13.4	14.4	15.9	17.9	18.3		
(21)		2% DB	38.3	35.7	32.6	26.4	20.3	15.9	14.8	17.3	22.4	28.4	32.7	35.7		
(22)		2% MCWB	19.2	19.1	18.3	15.9	13.7	11.6	11.0	11.9	14.2	15.3	17.1	17.7		
(23)		5% DB	35.5	33.2	30.3	24.7	18.6	14.7	13.7	15.7	20.0	25.4	30.0	32.8		
(24)		5% MCWB	18.8	18.9	17.8	15.3	13.2	11.2	10.5	11.2	13.5	14.8	16.6	17.5		
(25)		10% DB	32.9	31.2	28.0	22.7	17.0	13.5	12.7	14.2	18.0	22.8	27.4	30.1		
(26)		10% MCWB	18.4	18.3	17.1	14.7	12.6	10.7	9.9	10.4	12.6	14.0	16.2	17.0		
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	23.1	23.6	21.5	18.6	16.5	14.0	13.2	14.1	17.0	18.6	20.9	21.7		
(28)		0.4% MCDB	28.4	31.4	27.3	24.5	18.6	15.8	14.7	18.0	21.5	25.7	28.3	26.7		
(29)		2% WB	21.8	22.0	20.2	17.3	14.9	12.7	11.8	12.8	15.5	17.1	19.3	20.2		
(30)		2% MCDB	30.5	28.5	27.7	22.5	18.0	14.7	13.6	16.0	19.9	23.3	26.6	27.4		
(31)		5% WB	20.7	20.8	19.1	16.3	13.9	11.9	11.0	11.7	14.2	16.0	18.2	19.1		
(32)		5% MCDB	30.5	28.2	27.0	21.9	17.3	13.8	13.0	14.7	18.7	21.9	25.3	27.9		
(33)		10% WB	19.6	19.7	18.1	15.4	13.1	11.1	10.3	10.7	13.0	14.9	17.2	18.1		
(34)		10% MCDB	30.3	28.0	26.1	21.5	16.3	13.0	12.1	13.7	17.4	21.7	24.6	27.8		
(35)	Mean Daily Temperature Range	MDBR	15.6	14.6	13.8	13.1	11.2	9.6	8.9	10.4	12.1	13.9	14.6	15.2		
(36)		5% DB MCDBR	19.7	17.9	17.1	16.1	13.4	11.6	10.3	13.0	15.6	18.7	19.2	19.4		
(37)		5% WB MCWBR	5.6	5.4	5.7	7.1	7.6	7.6	6.9	8.1	8.3	7.9	6.5	6.0		
(38)		5% WB MCDBR	15.5	14.4	13.6	12.8	10.8	8.9	8.6	11.2	13.3	14.5	14.1	14.9		
(39)	Clear-Sky Solar Irradiance	taub	0.346	0.345	0.322	0.313	0.301	0.293	0.290	0.300	0.328	0.330	0.344	0.337		
(40)		taud	2.474	2.492	2.552	2.546	2.555	2.576	2.583	2.532	2.442	2.460	2.459	2.494		
(41)		Ebn at Noon	988	966	952	900	854	834	860	902	929	971	983	1002		
(42)		Edh at Noon	116	110	95	84	73	66	70	84	105	112	117	115		
(44)	All-Sky Solar Radiation	RadAvg	7.73	6.83	5.59	3.98	2.72	2.08	2.25	3.18	4.51	5.97	6.94	7.72		
(45)		RadStd	0.41	0.37	0.25	0.19	0.18	0.24	0.12	0.24	0.39	0.47	0.35	0.34		

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+116	(46)
(47)	Regional (31 neighbors)	+0.50	N/A	N/A	+0.95	N/A	N/A	-20	-118	+174	+71	(47)

Nomenclature: See separate page